

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 4 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------------|--|
| 1. Rational function D | (A) A value that makes the <i>original</i> denominator zero, so it can never be an input. |
| 2. Domain restriction A | (B) A horizontal line the graph approaches at its far left and far right; found by comparing the degrees of the numerator and denominator. |
| 3. Vertical asymptote H | (C) The number k in $y = kx$ or $y = \frac{k}{x}$; you find it from one known pair of values. |
| 4. Hole E | (D) A quotient of two polynomials whose denominator is not zero. |
| 5. Horizontal asymptote B | (E) A single missing point on an otherwise unbroken curve, left behind by a factor that cancels. |
| 6. Least common denominator G | (F) A number that turns up when you solve but must be thrown out, because it is an excluded value of the original equation. |
| 7. Extraneous solution F | (G) The smallest expression that every denominator divides evenly, built from the factored denominators. |
| 8. Constant of variation C | (H) A vertical line the graph runs alongside but never touches, left by a factor that does <i>not</i> cancel. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The domain of $f(x) = \frac{x+5}{x^2-9}$ is all real numbers except (A) $x = -5$ (B) $x = 3$ (C) $x = \pm 3$ ← (D) none — all reals
- Simplified completely, $\frac{x^2-25}{x^2+3x-10}$ is (A) $\frac{x-5}{x-2}, x \neq 2$ (B) $\frac{x-5}{x-2}, x \neq -5, 2$ ← (C) $\frac{x+5}{x-2}, x \neq 2$ (D) $\frac{-5}{3x-10}$
- The horizontal asymptote of $f(x) = \frac{3x^2-1}{x^2+4x}$ is (A) $y = 0$ (B) $y = \frac{1}{3}$ (C) $y = 3$ ← (D) there is none
- Which statement about $h(x) = \frac{x^2-x-6}{x^2-9}$ is true?
 - There are vertical asymptotes at $x = 3$ and at $x = -3$.
 - There is a hole at $x = 3$ and a vertical asymptote at $x = -3$.** ←
 - There is a hole at $x = -3$ and a vertical asymptote at $x = 3$.
 - There is a hole at $x = -2$ and a vertical asymptote at $x = 3$.

5. The graph of $g(x) = \frac{1}{x-4} + 2$ has asymptotes (A) $x = 4, y = 2 \leftarrow$ (B) $x = -4, y = 2$ (C) $x = 4, y = 0$
(D) $x = 2, y = 4$
6. y varies *inversely* as x , and $y = 8$ when $x = 3$. When $x = 12, y =$ (A) $2 \leftarrow$ (B) 6 (C) 24 (D) 32

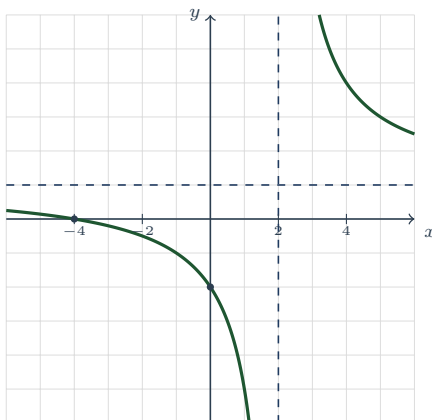
Teacher Note

Item 1: set the *denominator* to zero — $x^2 - 9 = 0$ gives $x = \pm 3$; (A) is the numerator-zero error and (B) keeps only one of the two. Item 2: $\frac{(x-5)(x+5)}{(x+5)(x-2)}$. (A) is the signature Unit 4 error — restrictions read off the *simplified* denominator, losing the hole at $x = -5$; (D) cancels the x^2 terms. Item 3: equal degrees, so the horizontal asymptote is the ratio of leading coefficients, $\frac{3}{1}$; (A) is the $n < m$ rule, (B) inverts the ratio. Item 4: $\frac{(x-3)(x+2)}{(x-3)(x+3)}$ — the $(x-3)$ cancels (hole), the $(x+3)$ does not (wall). (A) skips the cancel test entirely; (D) reads a restriction off the numerator. Item 5: $x - 4$ shifts *right*, so the wall moves to $x = 4$; $+2$ lifts the horizontal asymptote to $y = 2$. Item 6: $k = xy = 24$, so $y = \frac{24}{12} = 2$. (D) is the direct-variation answer, (C) reports k itself.

Part C — Short Answer & Computation (40 pts)

Show your work. State every restriction on the variable.

1. **Read the graph.** The rational function f is graphed below; the dashed lines are its asymptotes and each gridline is one unit.



- (a) Equation of the vertical asymptote: $x = 2$ Equation of the horizontal asymptote: $y = 1$
 (b) x -intercept: $(-4, 0)$; y -intercept: $(0, -2)$
 (c) Domain (interval notation): $(-\infty, 2) \cup (2, \infty)$
 (d) Range (interval notation): $(-\infty, 1) \cup (1, \infty)$
 (e) End behavior: as $x \rightarrow \pm\infty, f(x) \rightarrow 1$
 (f) Interval(s) where f is decreasing: $(-\infty, 2)$ and $(2, \infty)$ — **decreasing on each branch, but not on their union**
 (g) Interval(s) where $f(x) > 0$: $(-\infty, -4) \cup (2, \infty)$
2. **Simplify and state the restrictions.** Read the restrictions from the *original* denominator.

(a) $\frac{3x^2 - 27}{x^2 + x - 12}$

(b) $\frac{x^2 - 16}{4 - x}$

(a) **GCF first, then difference of squares:** $\frac{3(x-3)(x+3)}{(x+4)(x-3)} = \frac{3(x+3)}{x+4}, x \neq 3, x \neq -4$. The restriction $x \neq 3$ survives even though the factor cancelled — that is the hole.

(b) $4 - x = -(x - 4)$, so $\frac{(x-4)(x+4)}{-(x-4)} = -(x+4), x \neq 4$. Opposite binomials leave a factor of -1 behind.

3. **Multiply or divide.** Factor first; give the simplest form with its restrictions.

- (a) $\frac{x^2 - 4}{x^2 + 6x + 9} \cdot \frac{x + 3}{x - 2} = \frac{(x - 2)(x + 2)}{(x + 3)^2} \cdot \frac{x + 3}{x - 2} = \frac{x + 2}{x + 3}$, $x \neq -3$, $x \neq 2$. **Cancelling runs across the whole product — the $(x - 2)$ in the first numerator meets the $(x - 2)$ in the second denominator.**
- (b) $\frac{x^2 - x - 6}{x^2 - 1} \div \frac{x - 3}{x + 1}$ **Keep-change-flip:** $\frac{(x - 3)(x + 2)}{(x - 1)(x + 1)} \cdot \frac{x + 1}{x - 3} = \frac{x + 2}{x - 1}$, $x \neq 1$, $x \neq -1$, $x \neq 3$. **Three sources of restriction: the original denominator (± 1), the divisor's denominator (-1 again), and the divisor's *numerator* ($x = 3$, because it becomes a denominator after the flip).**

4. **Add, subtract, and simplify a complex fraction.**

- (a) $\frac{5}{x + 2} - \frac{3}{x - 1}$ **LCD** = $(x + 2)(x - 1)$: $\frac{5(x - 1) - 3(x + 2)}{(x + 2)(x - 1)} = \frac{5x - 5 - 3x - 6}{(x + 2)(x - 1)} = \frac{2x - 11}{(x + 2)(x - 1)}$, $x \neq -2$, $x \neq 1$. **The subtraction hits *both* terms of $3(x + 2)$.**
- (b) $\frac{x}{x^2 - 9} + \frac{2}{x + 3}$ **Factor first:** $x^2 - 9 = (x - 3)(x + 3)$, so the LCD is already $(x - 3)(x + 3)$: $\frac{x + 2(x - 3)}{(x - 3)(x + 3)} = \frac{3x - 6}{(x - 3)(x + 3)} = \frac{3(x - 2)}{(x - 3)(x + 3)}$, $x \neq \pm 3$. **Nothing cancels — $(x - 2)$ is not a factor of either denominator factor.**
- (c) $\frac{\frac{1}{x} + \frac{1}{2}}{\frac{1}{x} - \frac{1}{2}}$ **Combine top and bottom over the common denominator $2x$, then divide:** $\frac{\frac{2+x}{2x}}{\frac{2-x}{2x}} = \frac{2+x}{2-x}$, $x \neq 0$, $x \neq 2$. **The $x \neq 0$ comes from the little fractions and would be invisible in the final answer.**

5. **The parent function and its transformations.** Let $g(x) = \frac{1}{x + 3} - 2$.

- (a) Describe the transformation(s) that take $y = \frac{1}{x}$ to g . **left 3 units, down 2 units**
- (b) Vertical asymptote: $x = -3$; horizontal asymptote: $y = -2$
- (c) Domain: $x \neq -3$; range: $y \neq -2$
- (d) A different hyperbola of the form $y = \frac{1}{x - h} + k$ has vertical asymptote $x = 1$ and horizontal asymptote $y = 4$. Write its equation. $y = \frac{1}{x - 1} + 4$

6. **Analyze from the equation.** For $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$:

- (a) Factor the numerator and denominator, and list every excluded value. $f(x) = \frac{2(x - 2)(x + 2)}{(x - 3)(x + 2)}$. **The original denominator is zero at $x = 3$ and $x = -2$, so both are excluded.**
- (b) Which excluded value gives a *hole* and which gives a *vertical asymptote*? Give the coordinates of the hole. **$(x + 2)$ cancels, so $x = -2$ is a hole; $(x - 3)$ stays, so $x = 3$ is a vertical asymptote. Height of the hole from the simplified form $\frac{2(x - 2)}{x - 3}$ at $x = -2$: $\frac{2(-4)}{-5} = \frac{8}{5}$, so the hole is $(-2, \frac{8}{5})$.**
- (c) Horizontal asymptote: $y = 2$ (Justify with a degree comparison.) **Numerator and denominator are both degree 2, so the asymptote is the ratio of leading coefficients, $\frac{2}{1} = 2$.**
- (d) x -intercept: $(2, 0)$; y -intercept: $(0, \frac{4}{3})$

7. **Solve. Check every answer against the excluded values.**

- (a) $\frac{x}{x + 2} + \frac{1}{x - 1} = 1$ **Excluded:** $x \neq -2, 1$. **Multiply by the LCD $(x + 2)(x - 1)$:** $x(x - 1) + (x + 2) = (x + 2)(x - 1)$, so $x^2 + 2 = x^2 + x - 2$, giving $x = 4$. **Since 4 is not excluded, $x = 4$ is the solution.**
- (b) $\frac{x}{x - 4} = \frac{4}{x - 4} + 3$ **Excluded:** $x \neq 4$. **Multiply by $(x - 4)$:** $x = 4 + 3(x - 4) = 3x - 8$, so $-2x = -8$ and $x = 4$. **But 4 is exactly the excluded value — it is extraneous. The equation has no solution.**
- (c) $\frac{x}{x - 2} + \frac{2}{x} = \frac{4}{x^2 - 2x}$ **Factor the last denominator:** $x^2 - 2x = x(x - 2)$, so the LCD is $x(x - 2)$ and

the excluded values are $x \neq 0, 2$. Multiplying through: $x^2 + 2(x - 2) = 4$, so $x^2 + 2x - 8 = 0$ and $(x + 4)(x - 2) = 0$, giving $x = -4$ or $x = 2$. Throw out $x = 2$ (excluded). Only $x = -4$ survives; check: $\frac{-4}{-6} + \frac{2}{-4} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ and $\frac{4}{24} = \frac{1}{6}$ ✓

8. **Variation.** The table below records a pair of related quantities.

x	2	4	5	10
y	30	15	12	6

(a) Is y directly or inversely proportional to x ? Give the computation from the table that proves it.

Inversely — the products are constant: $2 \cdot 30 = 4 \cdot 15 = 5 \cdot 12 = 10 \cdot 6 = 60$ (the ratios y/x are not)

(b) Find the constant of variation k and write the equation. $k = 60$, so $y = \frac{60}{x}$

(c) Use your equation to find y when $x = 8$. $y = \frac{60}{8} = 7.5$

Teacher Note

Item 1(f) is the Unit 4 habit-breaker: the answer is two *separate* intervals, never $(-\infty, \infty)$ minus a point written as one interval — nothing crosses a wall. Item 1(d): accept “all reals except $y = 1$.” Items 2–4: a simplified form with *no* restriction list is half an answer — score it as such; the restriction must come from the *original* denominator. Item 3(b) is the three-sources item: watch for the missing $x \neq 3$ from the divisor’s numerator. Item 6(b): the hole’s height must come from the *simplified* form; substituting into the original gives $\frac{0}{0}$. Item 7(b) is the payoff loop — “ $x = 4$ ” as a final answer earns no credit; the check against the excluded values is the graded step. Accept “no solution” or “ $\emptyset$ ”. Item 8(a): a student who computes y/x and finds it changing has still shown the right work, provided they then check the products.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \frac{x^2 - 5x + 4}{x^2 - 16}$. (a) Factor both parts and list every excluded value of the *original* function. (b) Run the cancel test: which excluded value is a hole and which is a vertical asymptote? Give the hole’s coordinates and explain how you found its height. (c) State the horizontal asymptote, justify it with a degree comparison, and write the end behavior it describes. (d) Give the x - and y -intercepts. (e) Write the domain in interval notation. (f) A classmate simplifies f to $\frac{x-1}{x+4}$ and says the two functions are the same.

Say exactly where they agree and where they do not. (a) $f(x) = \frac{(x-1)(x-4)}{(x-4)(x+4)}$. **The original denominator is zero at $x = 4$ and $x = -4$, so both are excluded.**

(b) $(x-4)$ cancels, so $x = 4$ is a *hole*; $(x+4)$ does not, so $x = -4$ is a *vertical asymptote*. The hole’s height comes from the simplified form $\frac{x-1}{x+4}$ evaluated at $x = 4$: $\frac{3}{8}$. The hole is $(4, \frac{3}{8})$. Substituting 4 into the *original* would give $\frac{0}{0}$, which is why the simplified form is the one to use.

(c) Both degrees are 2, so the horizontal asymptote is the ratio of leading coefficients, $\frac{1}{1} = 1$: the line $y = 1$. That is the end-behavior statement — as $x \rightarrow -\infty$ and as $x \rightarrow +\infty$, $f(x) \rightarrow 1$.

(d) x -intercept: the simplified numerator is zero at $x = 1$, giving $(1, 0)$. y -intercept: $f(0) = \frac{4}{-16} = -\frac{1}{4}$, giving $(0, -\frac{1}{4})$.

(e) $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$.

(f) They agree at every input both accept — every x other than 4 and -4 . They do not agree at $x = 4$: the simplified expression happily returns $\frac{3}{8}$ there, but $f(4)$ does not exist. Two equivalent expressions can have different *domains*, so cancelling never restores an input.

2. **Modeling.** A club pays a one-time setup fee of \$180 to a screen printer, plus \$7 for each shirt printed. The

average cost per shirt for an order of n shirts is

$$A(n) = \frac{7n + 180}{n}.$$

(a) Explain why A is a rational function, and state the domain that makes sense for this situation. (b) Find $A(20)$ and $A(60)$, with units. What is happening to the average cost? (c) Rewrite $A(n)$ in the form $7 + \frac{180}{n}$ and use it to state the horizontal asymptote. What does that number mean about this order? (d) Find the order size for which the average cost is exactly \$9 per shirt. (e) Can the average cost ever be exactly \$7? Answer using the horizontal asymptote. **(a) It is a quotient of two polynomials, $7n + 180$ over n . Algebraically the only restriction is $n \neq 0$; in context n must be a positive whole number of shirts, so the sensible domain is $n = 1, 2, 3, \dots$ — an order of 0 shirts has no “per shirt” cost at all.**

(b) $A(20) = \frac{140 + 180}{20} = \frac{320}{20} = \16 per shirt; $A(60) = \frac{420 + 180}{60} = \frac{600}{60} = \10 per shirt. The average cost is *falling* as the order grows, because the one-time \$180 is being split among more shirts.

(c) Divide term by term: $A(n) = \frac{7n}{n} + \frac{180}{n} = 7 + \frac{180}{n}$. As $n \rightarrow \infty$ the second piece shrinks toward 0, so the horizontal asymptote is $y = 7$. That \$7 is the true printing cost of one shirt — the setup fee per shirt is the whole difference between the average cost and \$7.

(d) Solve $\frac{7n + 180}{n} = 9$. Multiply by n (allowed, since $n \neq 0$): $7n + 180 = 9n$, so $180 = 2n$ and $n = 90$ shirts.

(e) No. $A(n) = 7 + \frac{180}{n}$, and $\frac{180}{n}$ is positive for every positive n , so $A(n) > 7$ always. The graph approaches the line $y = 7$ but never reaches it — the setup fee never fully disappears, it only gets spread thinner.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) both factorings with *both* excluded values from the original denominator; 2 pts (b) hole at $x = 4$ and wall at $x = -4$, with the height $\frac{3}{8}$ taken from the simplified form (1 pt only if the two are named but the height is missing or computed from the original); 1 pt (c) $y = 1$ with the degree comparison, not just the answer; 1 pt (d) both intercepts; 1 pt (e) the three-piece union — a domain written as one interval, or missing the $x \neq 4$, loses it. (f) is the concept check: fold it into (b)/(e) credit if you prefer, but a student who says the two functions are simply “the same” has missed the unit’s central idea. D2: 1 pt (a) rational *and* a context domain of positive whole numbers; 1 pt (b) both values with dollar units and the observation that cost is falling; 2 pts (c) the rewrite *and* $y = 7$ interpreted as the real per-shirt printing cost (1 pt for $y = 7$ alone); 1 pt (d) $n = 90$; 1 pt (e) “no,” justified by $\frac{180}{n} > 0$ or by the asymptote never being reached. Do not accept “because it’s an asymptote” with no reason attached.